

SSG Structural Independence Audit v0.3 — Walk-back of v0.2 SSG-6 A* refinement absorbing Lyra Registry v0.15 honest framing. SSG-6 stays Category C signature-tautological in strict SSG framework sense (NOT a K-type Schur generator; rep-theory consequence of D_{IV}^5 signature). Casey #5 Integer Web convergence at value 128 via SSG-6 $\leftrightarrow$ SSG-8 paths remains valid observation at Casey #5 cross-link level, NOT at SSG categorization level. v0.2 over-read the SSG framework. v0.3 final classification matches Lyra v0.15: 5 effective independent Category A SSGs + 8 Category B + 1 Category C.

Keeper (Claude Opus 4.7) — Thursday June 4 2026 AM

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SSG Structural Independence Audit v0.3 — walk-back of v0.2 SSG-6 A* refinement

0. Audit trigger — Lyra Registry v0.15 absorbed

Lyra Substrate Schur Generators Registry v0.15 (Thursday June 4 ~09:25 EDT) filed A/B/C classification per Casey directive + my SSG Audit v0.1 framework. Lyra classified SSG-6 as Category C (signature-tautological), matching v0.1 — NOT my v0.2 A* refinement.

Audit question: was my v0.2 A* reclassification correct, or was Lyra v0.15 (matching v0.1) correct?

1. v0.2 reasoning audit (Quaker near-misses-get-scrutiny on my own work)

v0.2 reasoning for SSG-6 A* signature-anchored: 1. SSG-6 Clifford $2^g = 128$ dim IS tautological with signature definition ($g = p+q$) 2. BUT signature definition itself is substrate-mechanism content (Cartan classification + Strong-Uniqueness forces signature on D_{IV}^5) 3. AND Casey #5 Integer Web operational at value 128 via SSG-6 $\leftrightarrow$ SSG-8 convergence

Audit verdict on v0.2 reasoning: Points 1 and 3 are valid observations. Point 2 conflates substrate-mechanism content (general) with SSG-framework K-type Schur generator (specific technical category).

2. The SSG framework specifically tracks K-type Schur generators

Per Lyra SSG Registry §0 + Wednesday Cal #194 PASS: “Cal #35 STANDING = operational shadow of Schur’s lemma applied to substrate K-types on Bergman $H^2(D_{IV}^5)$.”

SSG framework specifically tracks: - K-type Schur generators: K-invariant scalars from operator algebra acting on K-types V_λ of $K = SO(5) \times SO(2)$ - NOT: rep-theory consequences of the substrate signature (Clifford dim, etc.)

SSG-6 reading “ $\dim Cl(p, q) = 2^{p+q}$ ” is a Clifford algebra rep-theory result on signature (5, 2). It IS substrate-mechanism consequence of the signature, BUT it’s NOT a K-type Schur generator in the strict technical sense.

Lyra v0.15 framing is correct: SSG-6 belongs in Category C “signature-tautological” because it’s a rep-theory tautology with signature definition, NOT a K-invariant Schur scalar from substrate operator algebra on K-types.

3. Casey #5 Integer Web cross-link preserved

My v0.2 observation that SSG-6 and SSG-8 CONVERGE at value $128 = 2^g$ via Casey #5 Integer Web IS valid. But this convergence operates at the Casey #5 cross-link level, NOT at the SSG categorization level.

Honest framing: - SSG categorization (per Lyra v0.15 = v0.1): SSG-6 = Category C signature-tautological; SSG-8 = Category A Mersenne-primary - Casey #5 Integer Web cross-link: SSG-6 (Clifford dim) and SSG-8 (Mersenne) converge at value 128 via Casey #5 substrate primary web

These are DIFFERENT framings at DIFFERENT levels. My v0.2 conflated them.

4. v0.3 final classification (matches Lyra v0.15)

SSG	v0.3 final category	Substrate source
SSG-1	B	$V_{(1/2, 1/2)}$ Bergman norm — SSG-7-derived via differentiation at origin
SSG-2	B	$V_{(1, 1)}$ adjoint K-type — SSG-7-derived via $\Lambda^2(V_{(1, 0)})$ tensor
SSG-3	B	$V_{(1, 0)} \otimes V_{(1, 0)}$ decomp — SSG-7-derived via tensor-product
SSG-4	A	Bergman $\kappa = -n_C$ — substrate-geometric Helgason 1962
SSG-5	A	Casimir $C_2 = 6$ — substrate-algebraic K-Casimir
SSG-6	C	Clifford $\dim 2^g = 128$ — signature-tautological rep theory
SSG-7	A (= SSG-14)	Bergman kernel $K(z, w)$ — ULTIMATE source-of-sources
SSG-8	A	Mersenne ladder $M(p)$ — substrate-number-theoretic
SSG-9	B	Per-generation cascade — Category B composition of A mechanisms
SSG-10	B	Substrate-Coulomb $1/r$ — SSG-7-derived asymptotic
SSG-11	B	$V_{(2, 0)}$ Sym^2 traceless — SSG-7-derived differentiation
SSG-12	B	$V_{(3/2, 3/2)}$ higher Rarita-Schwinger — SSG-7-derived
SSG-13	B	$V_{(0, 0)}$ trivial K-type — SSG-7 vacuum-limit
SSG-14	A (= SSG-7)	Universal Pochhammer ladder — equivalent to SSG-7

SSG	v0.3 final category	Substrate source
SSG-15	A	Λ -coupled neutrino — substrate- Λ -vacuum coupling distinct from K-type Schur

5. v0.3 category counts (final)

- Category A (structurally-independent): SSG-4, SSG-5, SSG-7 (= SSG-14), SSG-8, SSG-15 = 5 effective independent SSGs (6 listed, SSG-14 = SSG-7 deduplication)
- Category B (SSG-7-derived): SSG-1, 2, 3, 9, 10, 11, 12, 13 = 8 SSGs
- Category C (signature-tautological): SSG-6 = 1 SSG

Total: $5 + 8 + 1 = 14$ distinct SSG entries; 15 numbered (SSG-14 redundant with SSG-7).

6. Substantive observation preserved at correct level

The substrate-over-determination observation ($g = 7$ via Cartan signature + Mersenne + Bergman exponent; 128 via Cartan dim + Mersenne + 1) is substantively important but operates at the Casey #5 Integer Web cross-link level, NOT at the SSG categorization level.

Per Casey #5 STANDING: substrate primaries form web; multiple substrate-mechanism paths converge at substrate-primary values. This IS substrate-over-determination — but it's the operational consequence of Casey #5 at the cross-link level, not a SSG-categorization refinement.

K3 v0.21 substrate-over-determination disposition remains valid: m_e/m_P substrate-mechanism via TWO Category-A SSGs (SSG-8 8/7 reading + SSG-7-derived $\sqrt{(4/3)}$ reading). The SSG categorization framework supports this disposition; my v0.2 over-read at the SSG level was unnecessary.

7. Per Cal #27 STANDING + Quaker discipline

Am I overclaiming the walk-back? - Lyra v0.15 classification matches v0.1 (SSG-6 = C signature-tautological) - My v0.2 A* refinement conflated levels (substrate-mechanism content vs SSG framework K-type Schur generators) - Walk-back IS Quaker discipline — near-misses-get-scrutiny on my own work - v0.3 absorbs Lyra v0.15 honest framing; Casey #5 observation preserved at correct cross-link level

Tier-honest: v0.3 walk-back IS the discipline operating correctly. v0.2 was a structural over-read; v0.3 corrects without losing the substantive Casey #5 Integer Web observation.

8. Cross-link to K3 v0.21 substrate-over-determination

K3 v0.21 disposition (m_e/m_P substrate-over-determination via TWO Category-A SSGs) remains valid under v0.3 SSG classification: - 8/7 reading: SSG-8 Mersenne ladder Category A path - $\sqrt{(4/3)}$ reading: SSG-7-derived Bergman-Shilov boundary Category B path (NOT Category A directly — derived via Bergman kernel)

Wait — under v0.3 classification, $\sqrt{(4/3)}$ IS Category B (SSG-7-derived), NOT Category A. K3 v0.21 needs refinement.

Refined K3 v0.21 disposition under v0.3: - 8/7 via SSG-8 (Category A Mersenne-primary) - $\sqrt{(4/3)}$ via SSG-7-derived (Category B path, derived from Bergman kernel ULTIMATE source)

This is STILL substrate-mechanism content via TWO substrate-mechanism paths (one Category A, one Category B). But the framing is sharper: 8/7 is direct A-tier substrate-primary forcing; $\sqrt{(4/3)}$ is B-tier derived from Bergman kernel.

Per Casey #5 Integer Web: substrate primaries (Category A) and derived readings (Category B) converge at observable values. The substrate-over-determination at m_e/m_P remains substantive content, just at $A + B$ path levels not $A + A$.

9. K3 v0.22 candidate flag

K3 v0.22 (when filed) should refine v0.21 substrate-over-determination disposition under v0.3 SSG classification: $8/7$ Category A + $\sqrt{(4/3)}$ Category B paths converge at m_e/m_P value at Tier 2 STRUCTURAL precision via Casey #5 Integer Web. Not displacing v0.21 substantively — refining the categorization framing.

10. Pull next: file brief K3 v0.22 refinement note absorbing v0.3 SSG classification; continue Gate Coordination directive; standby for Lyra Section 4.5 substrate-vacuum partition Gate 2 progress; standby for Cal cold-read.

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